

Can News Content Predict Tomorrow's Move?

origin: five US AI stocks (NVDA / MSFT / META / AMZN / GOOGL, 2021–2025)

**News content beats the next-day-direction baseline,
challenging the semi-strong EMH.**

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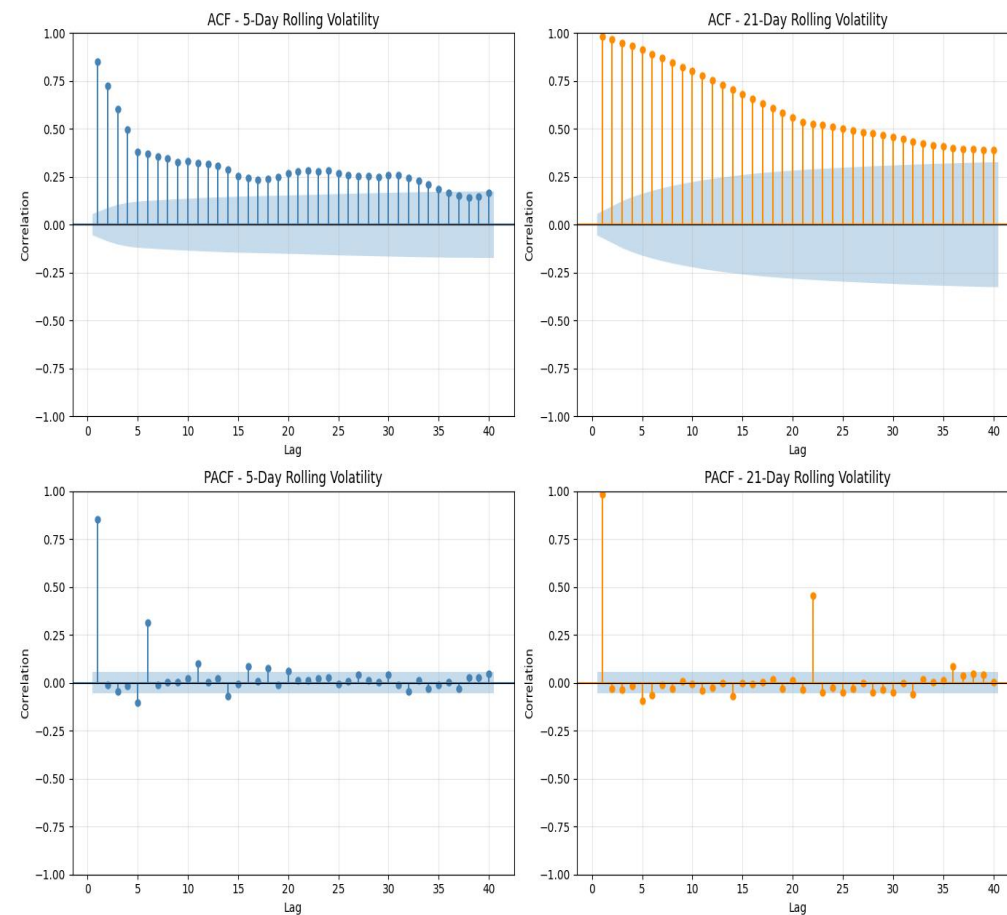
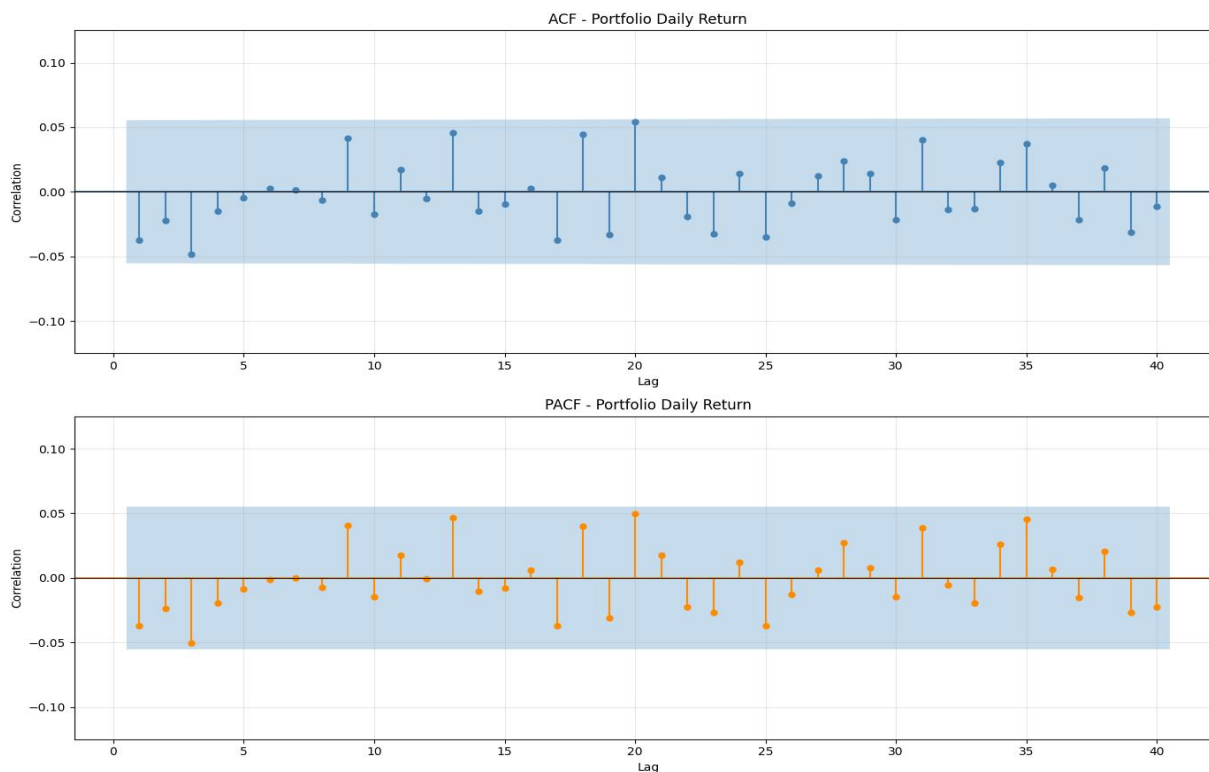




Data: Distribution & Autocorrelation

Stock	2021	2022	2023	2024	2025	Total
NVDA	60	72	335	829	1,138	2,434
MSFT	337	342	715	515	746	2,655
META	93	508	459	442	909	2,411
AMZN	1,032	810	710	608	822	3,982
GOOGL	193	207	312	315	597	1,624
Total	1,715	1,939	2,531	2,709	4,212	13,106
Share	13.1%	14.8%	19.3%	20.7%	32.1%	100%

13,106 Bloomberg articles · 5 stocks · 6,270 stock-days · 2025 alone = 32.1% of all news (skewed).

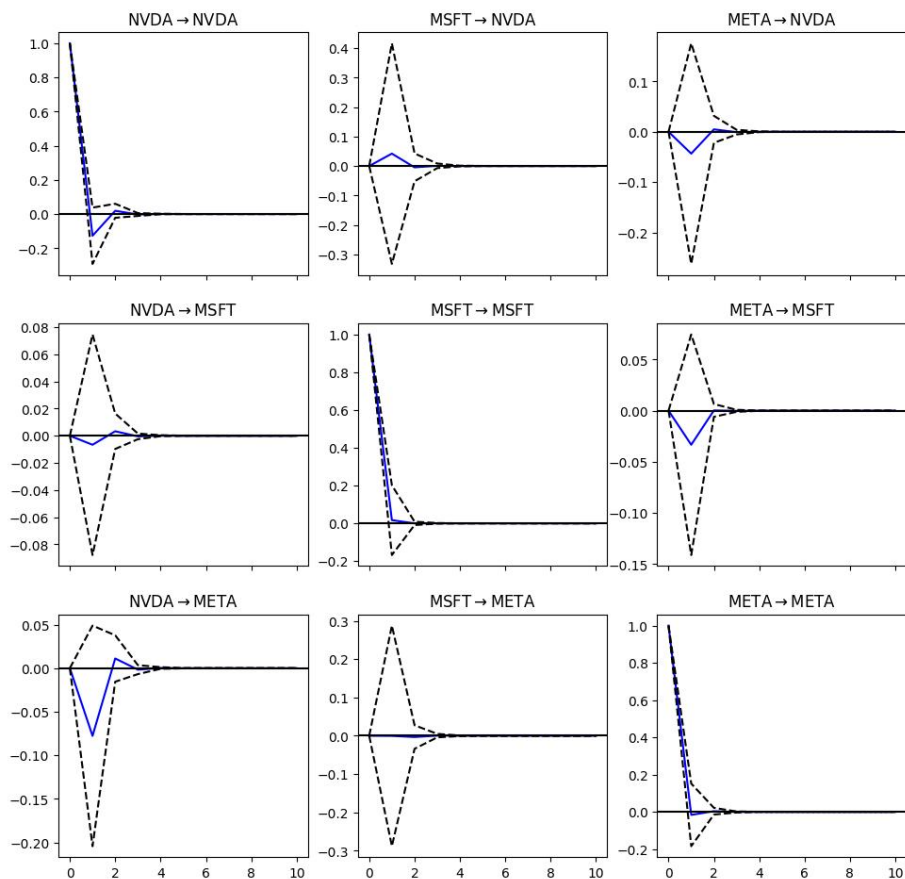


Returns show almost **no linear autocorrelation**

Volatility is **highly persistent** (clustering)

VAR + GSADF

IRF — 3-stock (no GOOGL/AMZN) close-close, 2025 (n=249, p=1)



VAR / IRF: cross-stock linkage is weak — each stock moves on its own info

GSADF: bubbles are stock-specific, not universal — only NVDA runs explosively (GSADF = 2.20, concentrated in 2024).





Background: BART + FinBERT

What they are

FinBERT — a BERT model fine-tuned on financial text; gives sentiment (positive/negative) + confidence, and reads finance wording correctly ("beat", "miss", "downgrade").

BART (zero-shot) — assigns each article to a narrative category from plain-language labels; no labeled training data needed.

Why these, not a dictionary or trained vectors

A **sentiment dictionary** just counts fixed positive/negative words — no context, misses finance meaning and negation.

Training our own word vectors / classifier isn't realistic — no labeled news, too few days per stock → it would just overfit. Pretrained transformers bring outside knowledge for free.

Categories are a modeling choice

The five categories (chip / power / algorithm / regulation / earnings) are defined by us on AI logic; BART tags each article zero-shot.

Different category definitions have different signals — so the classification is a design choice, not neutral (this is exactly what the Outlook wants to automate).



Failed Baselines

Method 1:

7-dim features (gated bart_* + FinBERT sign + confidence);

Bagging + RBF-SVM; train 2021–24 / test 2025 vs majority baseline.

Method 2 : double-0.6 gate; per-stock logit of direction on the 5 category signals; daily out-of-sample net-vote.

Per-ticker logistic regression (GLM, Binomial)

$$\Pr(\text{dir}_t = 1 \mid x_t) = \sigma\left(\beta_0 + \sum_{c \in \mathcal{C}} \beta_c x_{c,t}\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

$\mathcal{C} = \{\text{chip, power, algorithm, regulation, earnings}\}$; $x_{c,t} = \text{finbert_score}$ if the article passes the BART gate for category c , else 0.

Bagging Tree & SVM

Model	Train	Test 2025	Gap	Sig.
Bagging (200×d3)	0.60–0.67	0.45–0.55	+0.127	0 / 15
RBF-SVM	—	≈ baseline	+0.002 *	0 / 15

** 0 of 30 total significant; SVM gap tiny only because it predicts near-constantly.*

GLM Logistic Regression (2025)

Ticker	Test acc	Baseline	95% CI
NVDA	0.505	0.541	[0.414, 0.595]
MSFT	0.608	0.541	[0.500, 0.716]
META	0.506	0.519	[0.392, 0.620]
AMZN	0.547	0.516	[0.422, 0.672]
GOOGL	0.500	0.568	[0.392, 0.608]

Summary

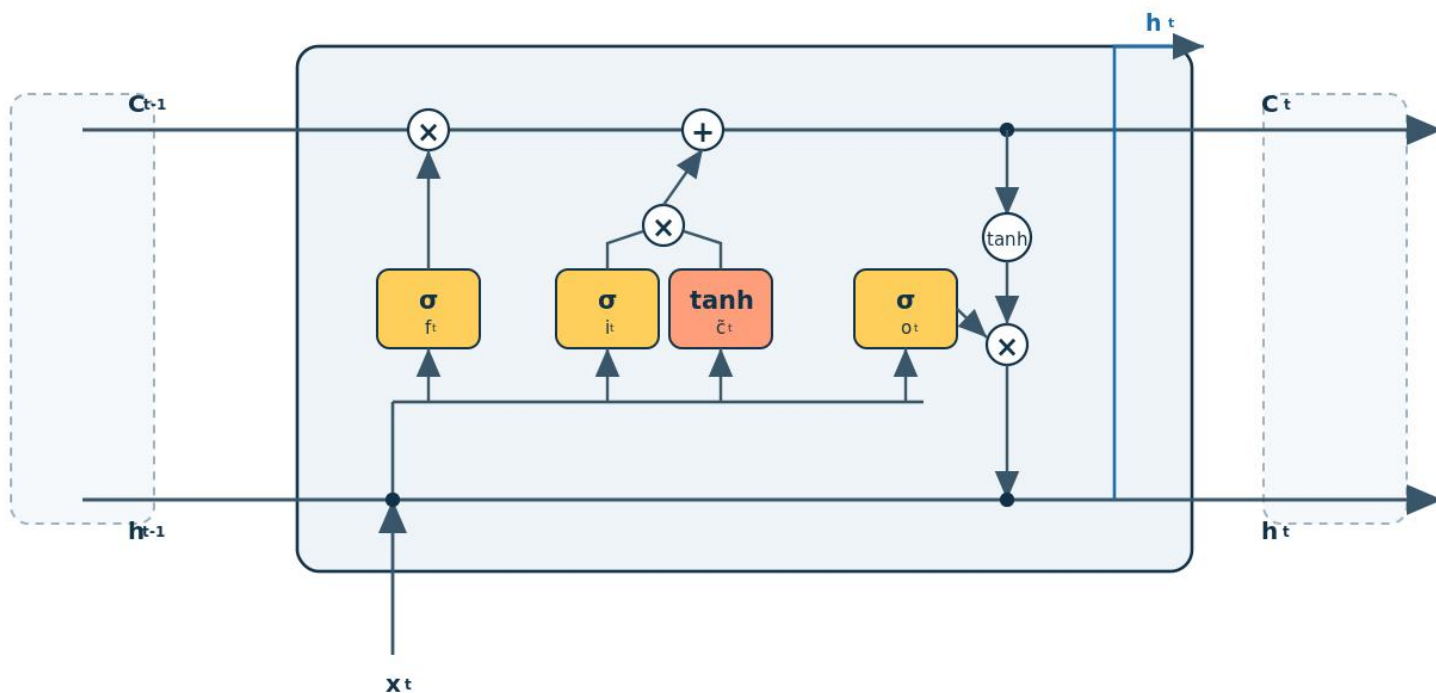
Whatever the model, flexible or linear, a single global model doesn't beat the baseline out-of-sample: trees/SVM overfit, logit only faint.

Failed Baselines

LSTM cell — propagating a $K = 10$ news-day window

Input per step: 6 features (finbert_score + 5 bart_*) + 5 one-hot ticker = 11-dim

Hidden size = 16 · window $K = 10$ · split 2021-23 train / 2024 val / 2025 test



Method 3 — Does news need memory?

An LSTM (16 units) reads the last 10 days of the 11-dim feature to predict next-day up/down;

train 2021–23, early-stop 2024, test 2025.

Summary— a bigger model just overfits more, not better. With so little data, the fix is gating, not a bigger model.

After the last step, the final hidden state is read out:



Result (2025, out-of-sample)

test 0.522 < baseline 0.533

train 0.631 → ≈11-pt gap: overfits harder



Failed Baselines

One linear regression per (stock, category), then vote

$$r_{t+1} = \alpha + \beta s_{c,t} + \gamma r_t + \varepsilon_t, \quad \widehat{\text{dir}}_{t+1}^{(c)} = \mathbb{1}[\hat{\beta} s_{c,t} > 0]$$

$s_{c,t}$ = daily mean finbert_score of category c (double-0.6 gate); daily prediction = majority vote of active categories.

FinBERT + LP, NO gating — first excess, but unreliable (2025)

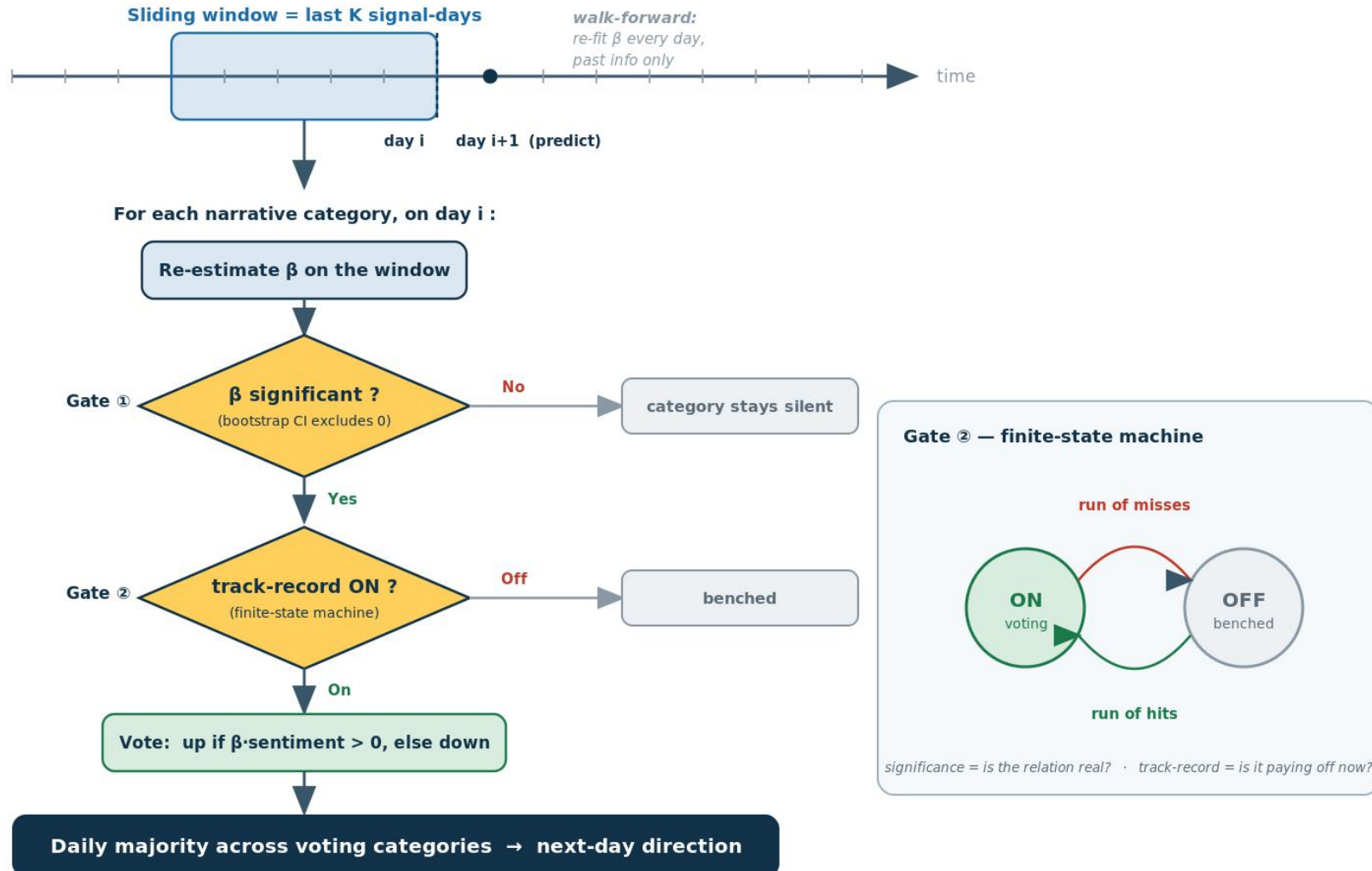
Stock	Acc	Base '24	Excess	Days	P>base
NVDA	61.5%	56.0%	+5.5%	96	88%
MSFT	58.3%	53.6%	+4.8%	60	73%
META	53.0%	53.2%	-0.1%	66	45%
AMZN	49.2%	53.2%	-4.0%	59	26%
GOOGL	45.7%	59.5%	-13.8%	70	1%

Method 4 — For each (stock, category), regress next-day return on that category's sentiment;

the sign of β says up or down. Each day the active categories vote, majority wins — β fixed on 2021–24, no gating.

Summary — For the first time, the signal **beats** the baseline (NVDA, MSFT). But it's **not stable**: other stocks are flat or negative.

NG-LP — sliding window + double gate



NG-LP = Narrative-Gated Local Projections

Local Projection — one simple regression: predict next-day return from a narrative's sentiment.

Narrative-gated — a narrative votes only when it's both real (Gate ①) and working (Gate ②).

Combine everything by **daily majority vote**.

NG-LP by stock — 2025 (out-of-sample)

Universe	Acc	Base	Excess	n	P>base
NVDA	63.3%	53.6%	+9.7%	49	91%
MSFT	66.7%	52.2%	+14.5%	27	92%
META	64.5%	51.8%	+12.8%	31	91%
AMZN	66.7%	50.9%	+15.7%	3 *	75%
GOOGL	40.0%	53.7%	-13.7%	25	9%
All 5 pooled	60.0%	52.9%	+7.1%	135	96%
CORE-3 pooled	64.5%	53.5%	+11.0%	107	99%

* AMZN has only 3 voting days — too few to trust. Weak-AI GOOGL is the control that fails (-13.7%).

Volatility stratification — CORE-3 pooled (U-shape)

Volatility band	n	Acc	95% CI	P>base	mean ret
Small	36	72.2%	[58.3%, 86.1%]	99%	0.26%
Mid	35	48.6%	[31.4%, 65.7%]	31%	1.05%
Large	36	72.2%	[55.6%, 86.1%]	99%	3.68%

Accurate on both quiet & shock days (72.2%), near-random in the middle (48.6%). Bands defined ex-post.

Results — On CORE-3 (NVDA/MSFT/META), NG-LP clearly beats the baseline, and every core stock beats its own ($P > \text{base} \geq 90\%$).

Control holds — the weak-AI stock GOOGL fails; dropping the weak-AI pair lifts the edge, the signal is specific to the AI narrative, not a market-wide effect.

Volatility U-shape — best on quiet and shock days, near-random in the middle → the model only helps when the market has a clear direction.



Conclusion, Limitation & Outlook

Conclusion

NG-LP beats each AI-core stock (+11.0 pts pooled, **64.5% vs 53.5%, P = 99%, out-of-sample**), news narrative carries next-day direction info, against a fully semi-strong-efficient market.

Why the simple model wins: with so **little data**, trees / SVM / LSTM all overfit and fall back to baseline, while a simple Local Projection actually holds up → the signal comes from gating, not from a bigger model.

Limitation

On 2026 data so far, NG-LP **fails to beat the baseline**, and the sample is still **too small to be significant** — so we can't confirm the narrative-gating logic holds beyond 2025; it may be specific to the 2025 market.

Outlook

Are the categories 2025-specific? Our five narrative categories are hand-picked, so the logic may only fit the 2025 narrative environment.

Toward self-selected narratives. Next step is a model that **discovers and selects its own narrative categories from the data**, instead of fixed hand-defined ones — so the gating logic generalizes across periods and stocks.